
Multi-Horizon Forecasting of Network Metrics Using Machine Learning

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 [GitHub Repo](#)

Abstract

Accurate forecasting of network metrics such as throughput and delay is essential for adaptive bitrate (ABR) algorithms, which rely on these predictions to make stable rate-selection decisions and maintain user Quality of Experience (QoE) during video streaming. While most prior work focuses on single-step forecasting, multi-horizon predictions can help ABRs anticipate future network changes rather than react to them. We study multi-horizon forecasting of key network metrics using real-world data, evaluating CatBoost, MLP, GRU, and LSTM models across several horizons. We also incorporate quantile regression via the pinball loss to capture uncertainty in highly variable network conditions. Our results show that tree-based models maintain the strongest point-forecast accuracy across horizons, while recurrent models offer competitive short-term performance. Quantile regression produces well-calibrated uncertainty intervals whose width grows appropriately with horizon, providing more informative predictions for downstream control.

1 Introduction

Video streaming now dominates more than 70% of global Internet traffic, driven by the growth of platforms such as YouTube, Twitch, and Netflix [Sidhu and Bentaleb \[2025\]](#). A key objective of these services is to maximize user Quality of Experience (QoE), which depends on delivering stable visual quality with minimal playback stalls [Barman and Martini \[2019\]](#).

Modern streaming systems rely on adaptive bitrate (ABR) algorithms that adjust playback quality according to fluctuating network conditions [Bentaleb et al. \[2018\]](#). When bandwidth drops, the ABR selects a lower bitrate to avoid rebuffering; when conditions improve, it increases quality [Sani et al. \[2017\]](#). Because these decisions depend heavily on future network behavior, accurate forecasting of throughput and delay is central to stable video streaming.

Since ABR is fundamentally a multi-step control problem, predictions are most useful when they provide visibility into several future horizons. Forecasts that capture these longer-range dynamics let the controller prepare for upcoming fluctuations rather than reacting after the fact, yielding more stable and higher-quality streaming [Saha et al. \[2024\]](#).

Despite this, prior work on network prediction focuses almost exclusively on single-step forecasts and typically reports only point errors such as Mean Average Error (MAE) [Yan et al. \[2020\]](#). This leaves open important questions about how accuracy changes with horizon length and how uncertainty estimates might long-horizon predictions.

In this work, we provide a systematic empirical study of multi-horizon network prediction using the Puffer dataset [Stanford Puffer Project \[2025\]](#). We evaluate both point forecasts and uncertainty-aware

forecasts across several model classes, with our implementation publicly available on Github¹. Our contributions are as follows:

1. We perform, to our knowledge, the first detailed empirical study of multi-horizon forecasting on the Puffer dataset, analyzing how prediction accuracy changes across horizons.
2. We compare four models, CatBoost, MLP, GRU, and LSTM, highlighting how different they perform under increasing horizons.
3. We incorporate quantile regression to produce prediction intervals, showing how uncertainty estimates become more informative as horizon length increases.

2 Literature Review

Network metric prediction has been a well-studied area in video streaming research, yet most prior work focuses exclusively on single-horizon forecasting of throughput and Round-Trip Time (RTT), reporting point metrics like MAE or MSE without characterizing how accuracy degrades across multiple future time steps or modeling prediction uncertainty [Sun et al. \[2016\]](#), [Yin et al. \[2021\]](#), [Narayanan et al. \[2020\]](#), [Nam et al. \[2021\]](#). Papers such as CS2P [Sun et al. \[2016\]](#), ANT [Yin et al. \[2021\]](#), Lumos5G [Narayanan et al. \[2020\]](#), and Xatu [Nam et al. \[2021\]](#) learn mappings from recent network traces to next-chunk estimates and integrate these into ABR controllers, but evaluate prediction performance only at one step-ahead, while also omitting systematic comparisons of tree-based, feedforward, and recurrent architectures as prediction horizons increase.

Puffer dataset studies have primarily targeted ABR policy learning and environment benchmarking rather than dedicated multi-horizon forecasting analysis of transport-layer metrics like delivery rate, and RTT [Stanford Puffer Project \[2025\]](#). Unlike this prior single-horizon focus, we conduct the a systematic study on Puffer comparing CatBoost, MLP, GRU, and LSTM across several horizons, while incorporating quantile regression to capture both point accuracy trends and calibrated uncertainty estimates over extended forecasts.

3 Methodology

Our goal is to evaluate the multi-horizon prediction accuracy of different machine learning models using the Puffer dataset. The idea is to have a model running on the client side that takes in observed network values and uses them to predict important metrics such as throughput (the amount of bytes recived in one second) and RTT (the delay from server to client)(Figure 1). This functions as a prediction module, and the output of this module can then be passed to a separate connection or quality optimization component containing an ABR algorithm. For example, this could be an RL-based controller which, by observing the current system state, selects the best quality level for the next video chunk, as shown in several prior works [Mao et al. \[2017\]](#), [Bentaleb et al. \[2024\]](#). In this paper, however, our focus is strictly on building and evaluating the prediction module itself. To this end, this section describes the data cleaning process, the applied machine learning models, and the evaluation metrics used in our study.

3.1 Data Pre-Processing

The dataset was downloaded from the Puffer data repository [Stanford Puffer Project \[2025\]](#). To decrease training time, only one day of data (February 10–11, 2025) was used for this stage. First, the timestamp column was converted from nanoseconds to seconds relative to the start of each session. Since the dataset contains multiple streaming sessions identified by unique `session_id` values, the data was grouped accordingly to preserve session-level structure for subsequent analysis. From this large dataset, we only take the first 800,000 rows and perform an 80/20 split for training and test data. Furthermore, we smooth the dataset using a 10-sample sliding window, a standard procedure to reduce noise in raw network data for better learnability [Zeng et al. \[2023\]](#).

Table 1 summarizes the main features derived from the Puffer dataset that will be used for model training and evaluation. These features were selected to capture both network-level and client-side characteristics relevant to Quality of Experience (QoE) forecasting.

¹[GitHub Repository](#)

Table 1: Extracted Features from the Puffer Dataset for Forecasting

Feature	Description
cwnd	Congestion window size (in number of packets) from TCP stack.
rtt	Delay time between client and server (μ s).
delivery_rate	TCP sender delivery rate estimate (bytes/s).
buffer	Playback buffer size at the client (seconds).

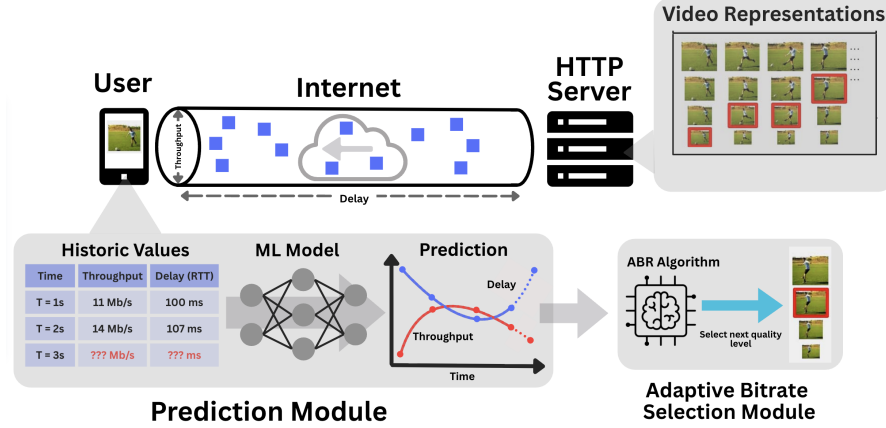


Figure 1: Simplified video streaming architecture; showcasing the importance of accurate throughput and RTT prediction for a smooth streaming experience.

3.2 ML Models

We evaluate four representative architectures for time-series prediction: CatBoost [Prokhorenkova et al. \[2018\]](#), a gradient-boosted tree model that uses a fixed tree structure and efficient split selections, MLP [Singh and Banerjee \[2019\]](#), a stack of fully connected layers with nonlinear activations, GRU [Chung et al. \[2014\]](#), a recurrent architecture that uses update and reset gates to control the flow of temporal information, and LSTM [Van Houdt et al. \[2020\]](#), a more advanced recurrent model with a dedicated cell state and input, output, and forget gates. Full architectural and training details are provided in Appendix A.

3.3 Evaluation Metrics

Our study includes both point forecasting and uncertainty-aware forecasting, which require different training objectives and evaluation criteria. For point predictions, models are trained and evaluated using MAE.

For quantile forecasting, the goal is to estimate conditional quantiles rather than a single future value. This is done using the *pinball loss* (also known as the Gumball loss), which applies asymmetric penalties to over- and under-estimation [Koenker and Bassett Jr \[1978\]](#). For a quantile level $\tau \in (0, 1)$, the loss is:

$$L_{\tau}(y, \hat{y}) = \begin{cases} \tau(y - \hat{y}), & y \geq \hat{y}, \\ (\tau - 1)(y - \hat{y}), & y < \hat{y}. \end{cases}$$

This objective enables models to learn calibrated quantile estimates and, by extension, prediction intervals. Quantile forecasts are evaluated in terms of *coverage* (the fraction of true values falling inside a predicted interval) and *interval width* (the sharpness of the interval). Formal metric definitions appear in Appendix B.

4 Experimental Results

4.1 Regression

We evaluate point forecasting performance for the two primary network metrics in our study: throughput (delivery rate) and RTT (delay). Each model predicts future values at horizons of 1, 2, 4, 8, and 16 steps, allowing us to quantify how accuracy deteriorates as temporal distance increases. Figure 2 illustrates model predictions for RTT at short (2-step) and long (16-step) horizons. At small horizons, most models track the underlying signal closely, with GRU showing slightly larger deviations. By contrast, at 16 steps ahead, all models struggle to reproduce rapid spikes and drops, smoothing out much of the fine-grained temporal structure. Additional prediction plots for both RTT and throughput across all horizons (1–16) appear in Appendix C.

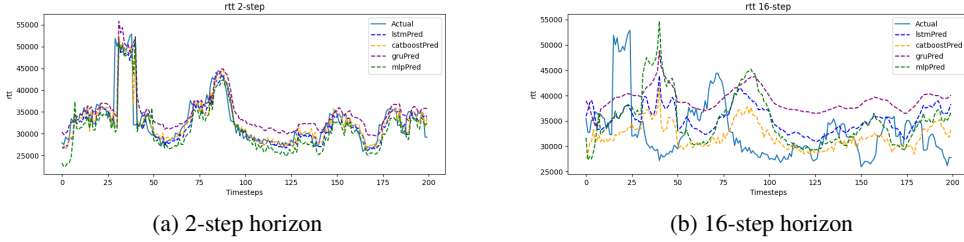


Figure 2: RTT predictions versus ground truth for short and long prediction horizons.

Figure 3 summarizes how test MAE grows with horizon for both metrics. As expected, error increases substantially with horizon length: MAE roughly triples between 1- and 16-step forecasts. CatBoost consistently achieves the lowest error across RTT horizons, with its advantage widening at longer steps. LSTM shows a slight edge at the 1-step RTT horizon but loses this advantage quickly as the horizon increases. GRU exhibits the largest degradation for both metrics, especially beyond 4 steps. For throughput, all models degrade at a similar rate, though CatBoost and MLP remain slightly stronger at long horizons. These trends highlight that tree-based models can remain competitive for short- and mid-range prediction.

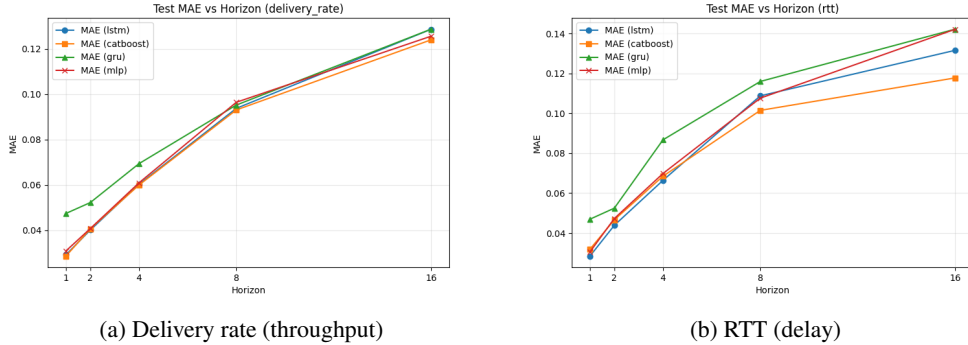


Figure 3: Model MAE as a function of prediction horizon for throughput and RTT.

4.2 Quantile Regression

Given the rapid degradation of point forecasts at longer horizons, we now evaluate models under quantile regression, where the goal is to predict an interval of plausible future values rather than a single point estimate. This is often more informative in networking scenarios, as systems such as ABR controllers benefit from knowing the range of likely outcomes. To estimate these ranges, we train all four model classes using the pinball loss to produce 25–75% quantile predictions at horizons of 1, 4, and 16 future steps.

Figure 4 shows example LSTM quantile predictions across these horizons. As expected, the uncertainty band widens as the prediction horizon increases: at 1-step, the model tracks RTT closely,

while at 16 steps the interval expands noticeably. Even so, the model continues to capture the overall trend, though sharp spikes and sudden drops remain difficult to anticipate. Appendix D contains the corresponding plots for CatBoost, GRU, and MLP.

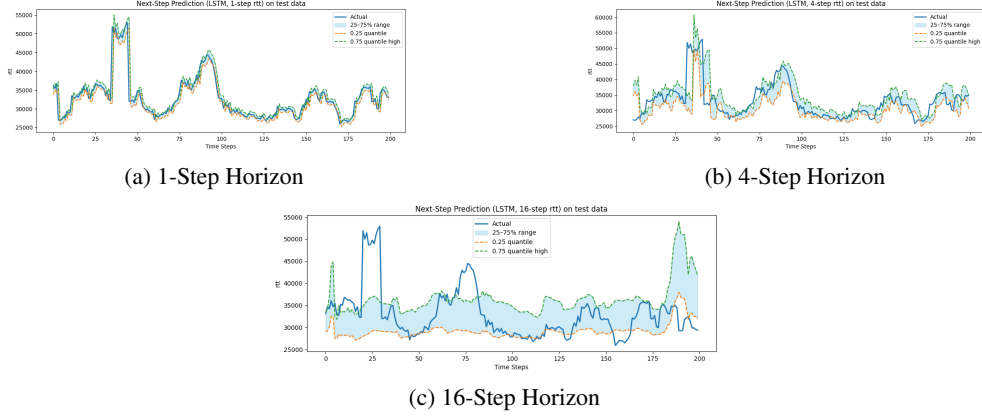


Figure 4: LSTM 25–75% quantile predictions at increasing horizons.

To compare models, Figure 5 plots their coverage (fraction of true values lying inside the predicted band) versus average interval width. The ideal model lies closest to the lower-right corner: high coverage with narrow uncertainty intervals.

For delivery rate at the 1-step horizon, LSTM performs best, achieving the highest coverage ($\approx 52\%$) while also producing the narrowest interval ($\approx 4.59 \times 10^5$). MLP attains only $\approx 39\%$ coverage, and GRU produces the widest interval ($\approx 4.92 \times 10^5$). At the 16-step horizon, performance shifts: LSTM still provides the narrowest interval ($\approx 1.85 \times 10^6$), while GRU achieves the best coverage ($\approx 46\%$). CatBoost and MLP sit between these extremes, offering a more balanced trade-off.

For RTT, the trends slightly differ. At the 1-step horizon, LSTM again shows the best balance with $\approx 60\%$ coverage and a low interval width (≈ 3000). GRU performs worst with $\approx 33\%$ coverage and a substantially wider interval. At 16 steps, all models degrade: coverage falls to the 30–36% range, and interval widths increase sharply. LSTM nevertheless yields the narrowest band, while CatBoost and MLP provide slightly higher coverage.

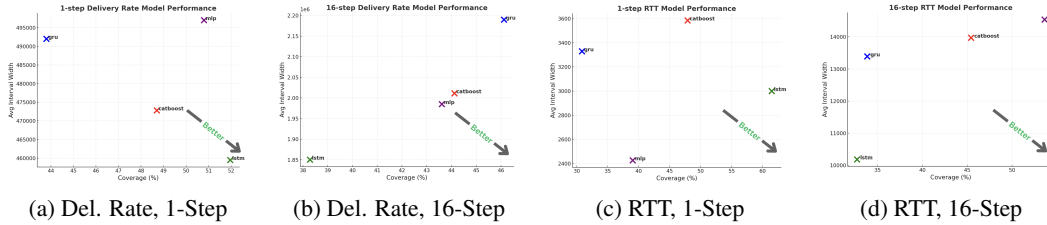


Figure 5: Coverage vs. interval width for 25–75% quantile predictions. Lower-right is better.

5 Concluding Remarks

In this project, we evaluated four ML models for multi-horizon forecasting of RTT and delivery rate. CatBoost performed best for point prediction, though all models showed substantial error growth as the horizon increased. Quantile regression provided a more informative view at longer horizons: LSTM had the best coverage–width balance at short horizons, but this advantage faded as the horizon grew and model performance converged, even though interval forecasts still captured overall trends better than point estimates. This work focused solely on forecasting and did not evaluate downstream effects on ABR decision-making; future work could integrate these predictions into real controllers, study how uncertainty estimates impact QoE, and explore lighter or more energy-efficient models for deployment on client devices.

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A Model Architectures and Hyperparameters

This appendix summarizes the architectural details and training configurations used for each model class. All experiments use identical train–test splits and preprocessing pipelines.

- **CatBoost**
 - Iterations: 2000
 - Learning rate: 0.05
 - Tree depth: 8
 - Subsample ratio: 0.8
 - Loss functions: MAE (point prediction), Quantile loss (quantile regression)
- **MLP**
 - Hidden layers: (128, 64, 32)
 - Activation: ReLU
 - Optimizer: Adam
 - Learning rate: 0.0005 (point regression); 0.001 (quantile regression)
 - Training epochs: 80–300 depending on task
 - Early stopping: Enabled (validation fraction 0.1)
- **LSTM**
 - Hidden dimension: 128–256
 - Number of layers: 3–4
 - Dropout: 0.1–0.2
 - Optimizer: AdamW
 - Losses: Huber loss (point prediction), Quantile Huber loss (quantile regression)
 - Gradient clipping: Enabled (max-norm 5)
- **GRU**
 - Hidden dimension: 128–256
 - Number of layers: 3–4
 - Dropout: 0.1–0.2
 - Fully connected head: $512 \rightarrow 256 \rightarrow 128 \rightarrow \text{output}$
 - Optimizer: AdamW
 - Losses: MSE (point prediction), Pinball loss (quantile regression)
 - Gradient clipping: Enabled (max-norm 5)
- **General Settings**
 - Batch size: 256–1024 depending on model
 - Device: GPU when available
 - Early stopping: patience 5–10
 - Scaling: StandardScaler fitted on training split
- **Hardware & Environment**
 - CPU: Ryzen 5 5600X
 - GPU: NVIDIA RTX 3070 (8 GB VRAM)
 - RAM: 32 GB DDR4
 - Operating System: Windows 10

B Evaluation Metrics: Formal Definitions

This appendix summarizes the formal definitions of the metrics used for training and evaluating the forecasting models.

- **Mean Absolute Error (MAE)**

- Used as both a training loss (for certain point models) and an evaluation metric.
- Defined as:

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|.$$

- Measures average absolute prediction deviation.

- **Coefficient of Determination (R^2)**

- Used only for evaluating point forecasts.
- Defined as:

$$R^2 = 1 - \frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{\sum_{t=1}^n (y_t - \bar{y})^2}.$$

- Captures the proportion of variance explained by the model.

- **Pinball (Gumball) Loss for Quantile Regression**

- Used as the training loss for quantile models.
- Encourages the predictor to place the τ -quantile estimate at the correct location in the conditional distribution.
- Defined for quantile level $\tau \in (0, 1)$ as:

$$L_\tau(y, \hat{y}) = \begin{cases} \tau (y - \hat{y}), & y \geq \hat{y}, \\ (\tau - 1)(y - \hat{y}), & y < \hat{y}. \end{cases}$$

- Produces asymmetric penalties: underestimation is penalized more for large τ , and overestimation is penalized more for small τ .

- **Coverage and Interval Width**

- Used for evaluating quantile-based uncertainty estimates.
- Coverage:

$$\text{Coverage} = \frac{1}{n} \sum_{t=1}^n \mathbf{1} \left\{ \hat{y}_t^{(L)} \leq y_t \leq \hat{y}_t^{(H)} \right\}.$$

- Width:

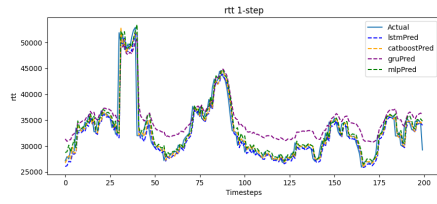
$$\text{Width} = \frac{1}{n} \sum_{t=1}^n \left(\hat{y}_t^{(H)} - \hat{y}_t^{(L)} \right).$$

- Together, they indicate calibration (correct probability mass) and sharpness (narrow intervals).

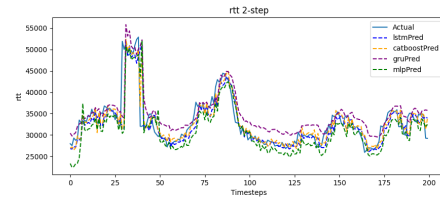
C Additional Plots: Point Predictions

This appendix includes the point-prediction outputs for RTT and delivery_rate at horizons 1, 2, 4, 8, and 16. Each row contains two horizons for compactness.

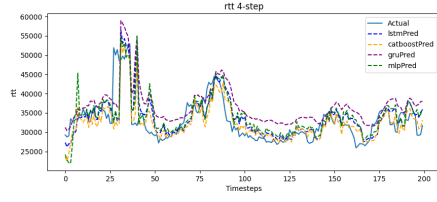
C.1 RTT Point Predictions



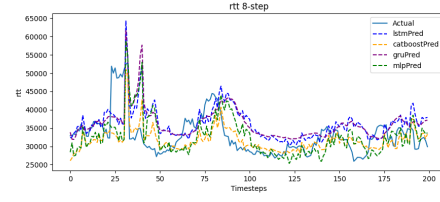
(a) RTT – 1-step



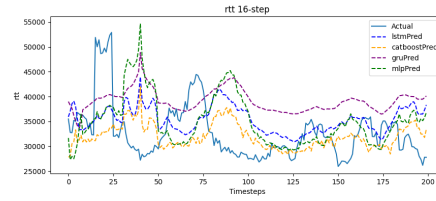
(b) RTT – 2-step



(c) RTT – 4-step



(d) RTT – 8-step



(e) RTT – 16-step

Figure 6: RTT point predictions across horizons.

C.2 Delivery Rate Point Predictions



Figure 7: Delivery rate point predictions across horizons.

D Additional Plots: Quantile Regression

This appendix provides the full 25–75% quantile regression plots for all models.

D.1 RTT Quantile Regression

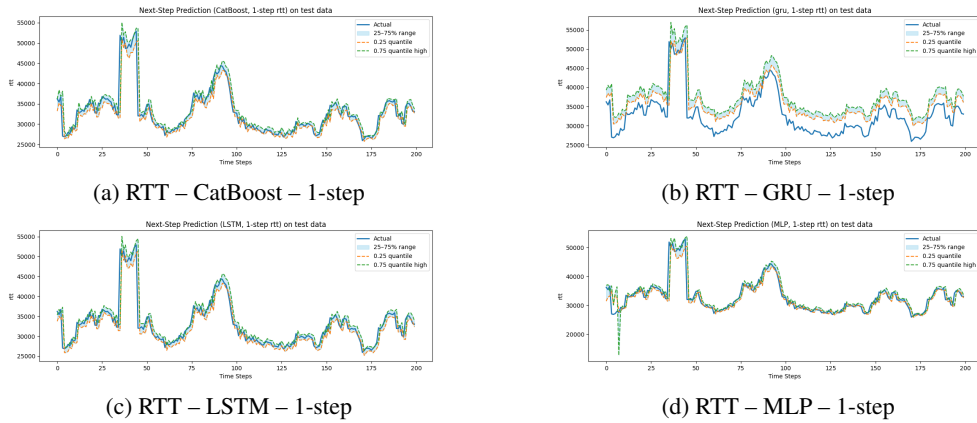
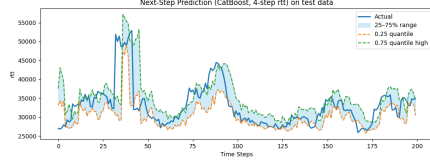
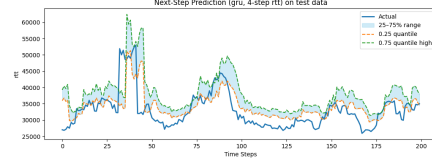


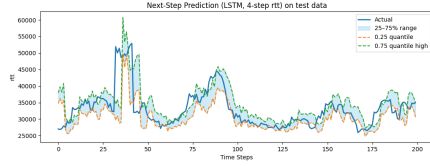
Figure 8: RTT 1-step quantile predictions.



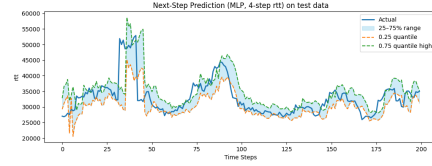
(a) RTT – CatBoost – 4-step



(b) RTT – GRU – 4-step

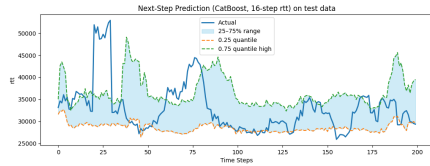


(c) RTT – LSTM – 4-step

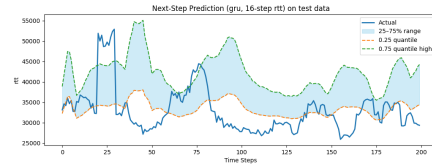


(d) RTT – MLP – 4-step

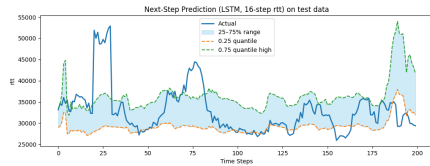
Figure 9: RTT 4-step quantile predictions.



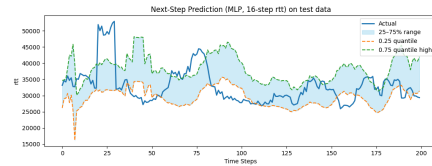
(a) RTT – CatBoost – 16-step



(b) RTT – GRU – 16-step



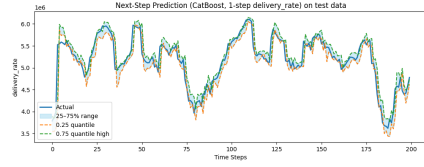
(c) RTT – LSTM – 16-step



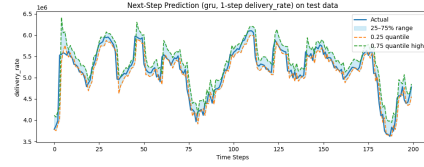
(d) RTT – MLP – 16-step

Figure 10: RTT 16-step quantile predictions.

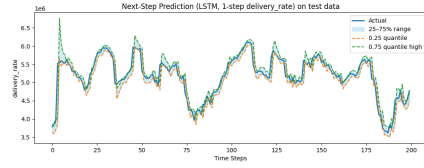
D.2 Delivery Rate Quantile Regression



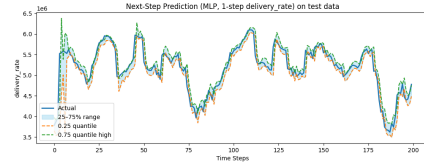
(a) Delivery Rate – CatBoost – 1-step



(b) Delivery Rate – GRU – 1-step

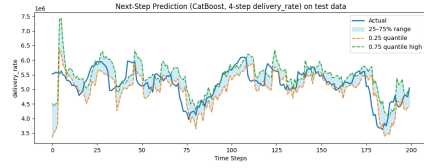


(c) Delivery Rate – LSTM – 1-step

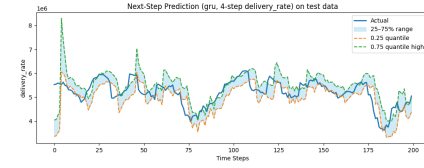


(d) Delivery Rate – MLP – 1-step

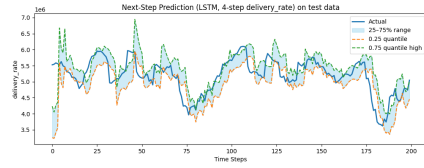
Figure 11: Delivery Rate 1-step quantile predictions.



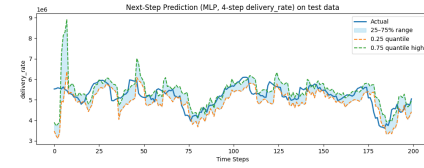
(a) Delivery Rate – CatBoost – 4-step



(b) Delivery Rate – GRU – 4-step

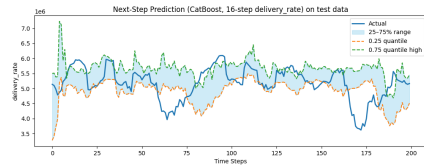


(c) Delivery Rate – LSTM – 4-step

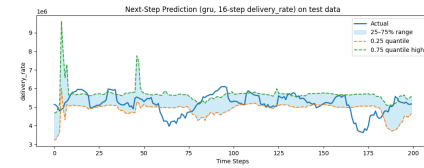


(d) Delivery Rate – MLP – 4-step

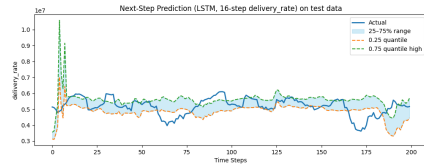
Figure 12: Delivery Rate 4-step quantile predictions.



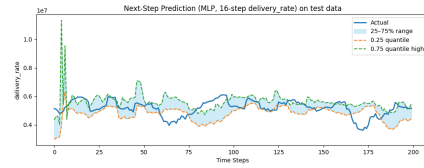
(a) Delivery Rate – CatBoost – 16-step



(b) Delivery Rate – GRU – 16-step



(c) Delivery Rate – LSTM – 16-step



(d) Delivery Rate – MLP – 16-step

Figure 13: Delivery Rate 16-step quantile predictions.

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